

Random variable: A variable that takes on certain values in accordance with specified probabilities.

Random event(E): failure of a component,

Random variable: Time to failure of the same component.

Event E occurs with some probability value $P(E)$ where $0 \leq P(E) \leq 1$.

Sample space: $S = \{E_1, E_2, E_3, \dots \dots \dots, E_k\}$ And $P(S) = 1$.

Complementary event of E is E^C

Hence $P(E) + P(E^C) = 1$

Mutually Exclusive - Occurrence of one precludes the occurrence of the other.

$$P(A \cup B) = P(A) + P(B) \text{ and } P(A \cap B) = P(\emptyset) = 0$$

A & B will be independent events when

$$P(A \cap B) = P(A) \times P(B).$$

When A & B are not independent the probability of their intersection must be defined using conditional probability.

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \rightarrow \text{equation of joint probability}$$

$$P(A|B) = \frac{P(A) \times P(B)}{P(B)} = P(A)$$

$$\begin{aligned}\text{Bayes' Formula: } P(A|B) &= \frac{P(A \cap B)}{P(B)} = \frac{P(B \cap A)}{P[(B \cap A) \cup (B \cap A^c)]} \\ &= \frac{P(B|A) \cdot P(A)}{P(B|A) \cdot P(A) + P(B|A^c) \cdot P(A^c)}\end{aligned}$$

Bayes' Formula in generalization

$$P(A_j|B) = \frac{P(B|A_j) \cdot P(A_j)}{\sum_{j=1}^n P(B|A_j) \cdot P(A_j)}$$

Random variable: It can be Discrete or Continuous

The probability distribution that assigns a probability to each value of a discrete random variable or assigns a probability to an interval of values of a continuous random variable can be described in terms of a probability mass function (PMF), $p(x)$ in the discrete case and a probability density function (PDF), $f(x)$ in the continuous case.

For both continuous and discrete distribution, a cumulative distribution function (CDF), $F(x)$ is defined. Both of these functions are commonly used in probability and statistics, and give a complete description of the probability distribution of a random variable.

The PMF or PDF describes the shape of the probability distribution., Whereas the CDF provides a cumulative probability i.e., $P\{X \leq x\} = F(x)$

DISCRETE DISTRIBUTIONS

For any discrete distribution we will define

$p(x) = \Pr\{X=x\}$ to be its PMF.

$F(x) = \Pr\{X \leq x\} = \sum p(x)$ is the CDF

Mean of the distribution $\mu = \sum_{all\ x} xp(x)$

variance of the distribution $\sigma^2 = \sum_{all\ x} (x - \mu)^2 p(x)$

Useful discrete distribution in reliability are Binomial and Poisson

BINOMIAL DISTRIBUTION.

Let X be the discrete random variable representing the no. of success in n independent trials. p is a constant and represents the prob. of success in each trial.

$X = 0, 1, 2, \dots, n$

The binomial PMF $p(x) = \binom{n}{x} p^x (1-p)^{(n-x)}$ where $x = 0, 1, 2, \dots, n$

Mean or expected value $E(X) = np$

Variance is $\text{Var}(X) = np(1-p)$.

Example: Let X be the discrete random variable representing the no. of failed components among five independent and identical components of which each component has 1 chance in 100 of failing. Then $X = 0, 1, \dots, 5$ and has a binomial distribution with $p = 0.01$ and $n = 5$. Find the mean no. of failures, variance and probability of exactly one failure.

POISSON DISTRIBUTION

X is a discrete random variable representing the no. of random occurrences (events) in a specified time; $X = 0, 1, 2, \dots \dots \dots$, the Poisson PMF is

$$p(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

Where $\lambda = E(X)$ is the mean no. of occurrences in a specified time and $\text{Var}(X) = \lambda$.

Example: Let X be the discrete random variable representing the no. of failures and subsequent repairs of a restorable system over a one-year period. Assuming X has a Poisson distribution with a mean of $\lambda = 2$ failures per year, calculate the prob. of not more than one failure in a year.

CONTINUOUS DISTRIBUTIONS

For a continuous random variable x , the PDF of x , is the function $f(x)$, for any numbers 'a' and 'b', that satisfy the equation:

$$P(a < x < b) = \int_a^b f(x)dx$$

For a random variable x , the CDF is the function $F(x)$, defined by:

$$F(X) = Pr(X \leq x) = \int_0^x f(x)dx$$

Which in other words, means that the cumulative distribution function is the probability that the value $X \leq x$.

The relationship between the PDF and the CDF is that the CDF is the cumulative values of the PDF, meaning that a point on the CDF function curve, is the area under the density function to the left of that point. Further, the PDF is the derivative of the CDF, which provide the following expression on the relationship between the PDF and the CDF:

$$f(x) = \frac{d(F(x))}{dx}$$

$$1. 0 \leq F(x) \leq 1$$

$$2. \Pr\{X \leq x\} = F(x) = \int_{-\infty}^x f(x)dx$$

$$3. f(x) = \frac{dF(x)}{dx}$$

$$4. \int_{-\infty}^{\infty} f(x)dx = 1$$

$$5. \Pr\{a \leq x \leq b\} = \int_a^b f(x)dx = F(b) - F(a)$$

$$6. E(x) = \mu = \int_{-\infty}^{\infty} xf(x)dx \quad \text{where } \mu \text{ is the mean or expected value of the random variable } x.$$

$$7. Var(x) = \sigma^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x)dx \quad \text{where } \sigma^2 \text{ is the variance of the distribution}$$

$$8. \text{ If } y = \sum_{i=1}^n a_i x_i \text{ where all the } x_i \text{ are independent random variables having means } \mu_i \text{ and variances}$$

$$\sigma_i^2 \text{ and } a_i \text{ are constants, then } E(y) = \sum_{i=1}^n a_i \mu_i \quad \text{and} \quad Var(y) = \sum_{i=1}^n a_i^2 \sigma_i^2$$

Reliability Function

Reliability is defined as the probability that a component or system will function over some time period t . To express the relationship mathematically, we define the continuous random variable T to be the Time to Failure (TTF) of the component or system and $T \geq 0$.

Then reliability can be expressed as

$$R(t) = \Pr\{T \geq t\}$$

$$\text{Where } R(t) \geq 0, R(0) = 1 \text{ and } \lim_{t \rightarrow \infty} R(t) = 0$$

Cumulative failure distribution Function

If we define $F(t)$, the probability that a failure occurs before time t ,

$$F(t) = 1 - R(t) = \Pr\{T < t\}$$

$$\text{Where } F(0) = 0 \text{ and } \lim_{t \rightarrow \infty} F(t) = 1$$

Failure Probability Density Function

$$\text{Probability density Function (PDF), } f(t) = \frac{d(F(t))}{dt} = -\frac{d(R(t))}{dt}$$

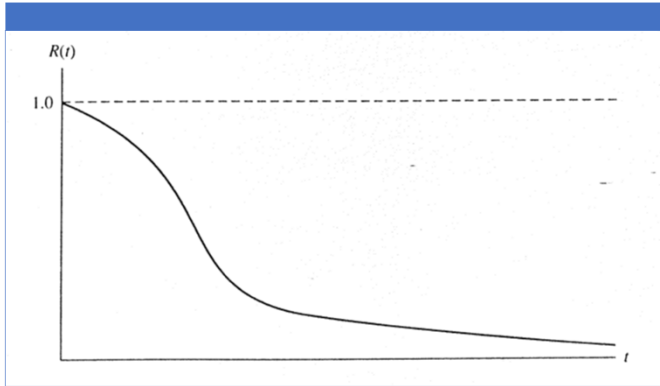
$$\int -d(R(t)) = \int f(t)dt$$

When t is t, Reliability is R(t) and when t is ∞ , reliability is 0, then

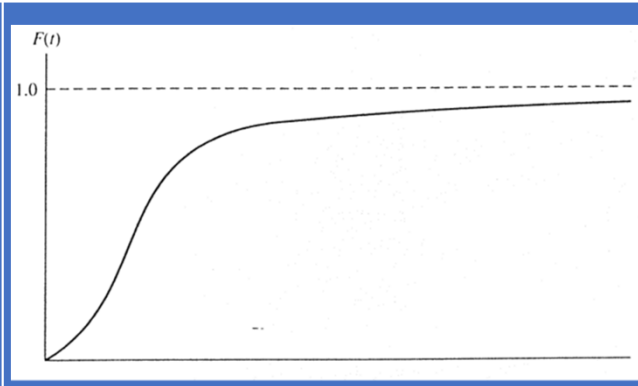
$$\int_{R(t)}^0 -d(R(t)) = \int_t^{\infty} f(t)dt$$

$$-R(t)\Big|_{R(t)}^0 = \int_t^{\infty} f(t)dt$$

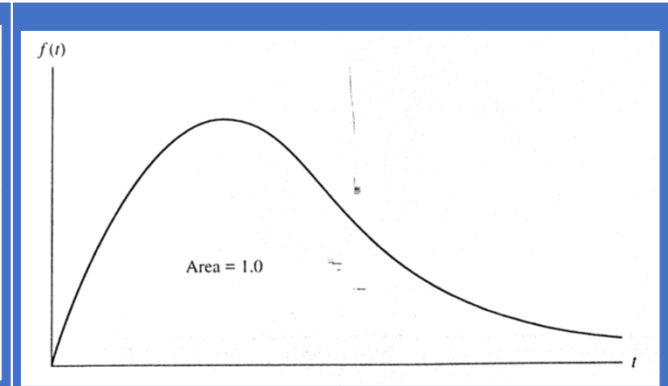
$$R(t) = \int_t^{\infty} f(t)dt$$



Reliability function



Cumulative distribution function



Probability density function

- Reliability function, $R(t)$
- Cumulative distribution function (CDF), $F(t)$
- Probability density function (PDF), $f(t)$
- Hazard rate function, $\lambda(t)$

The PDF describes the shape of the failure distribution and has the following properties.

$$f(t) \geq 0 \quad \text{and} \quad \int_0^{\infty} f(t)dt = 1$$

For a given PDF,

$$F(t) = \int_0^t f(t)dt \quad \text{and} \quad R(t) = \int_t^{\infty} f(t)dt$$

Example: Given the following PDF for the random variable T, the time (in operating hours) to failure of a compressor, give the expression for *the reliability function and the cumulative failure distribution function*. What is its reliability for a 100-hr operating life?

$$PDF = f(t) = \begin{cases} \frac{0.001}{(0.001t + 1)^2} & t \geq 0 \\ 0 & otherwise \end{cases}$$

Various summary measures of failure distribution used in reliability analysis

$$\text{Mean Time to Failure (MTTF)} = E(T) = \int_0^{\infty} t f(t) dt$$

$$= \int_0^{\infty} t \cdot \frac{-d(R(t))}{dt} dt = \int_0^{\infty} -t \cdot d(R(t))$$

Using integration by parts we get,

$$= t \cdot R(t) \Big|_0^{\infty} + \int_0^{\infty} R(t) dt = \int_0^{\infty} R(t) dt$$

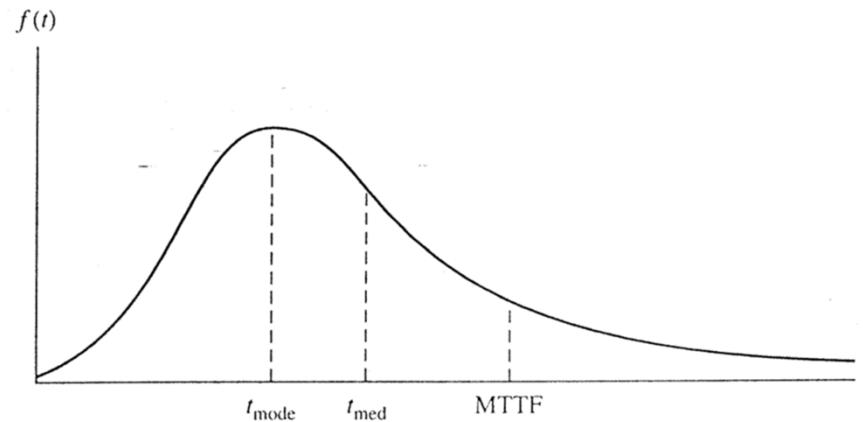
Median time to failure: $R(t_{med}) = 0.5 = \Pr(T \geq t_{med})$

The median may be preferred to mean when the distribution is highly skewed.

Mode: most likely observed time is

$$f(t_{mod}) = \max_{0 \leq t \leq \infty} f(t)$$

For a small fixed interval of time centered around the mode, the probability of failure will generally be greater than for an interval of the same size located elsewhere within the distribution.



Variance of time to failure

MTTF alone cannot uniquely characterize a failure distribution. One measure that is often used to further describe a failure distribution is its variance σ^2 , defined by

$$\sigma^2 = \int_0^{\infty} (t - MTTF)^2 f(t) dt$$

It is a measure of the spread, or dispersion, of the failure times about the mean. The variance can also be written as

$$\sigma^2 = \int_0^{\infty} t^2 f(t) dt - (MTTF)^2$$

It is computationally simpler than the above one. The standard deviation is easier to interpret than the variance as it is measure in the same time units as random variable (T)

Example: Consider the probability density function

$$f(t) = \begin{cases} 0.002e^{-0.002t} & t \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

When t is in operating hours. Calculate the MTTF, median time to failure and mode.

Hazard Rate Function

A hazard is a source of potential damage, harm or adverse effects on something or someone. Harm is the physical injury or damage to health. A potential source of harm to a worker is hazard.

Failure rate or hazard rate function, is often used in reliability. It provides an instantaneous (at time t) rate of failure.

$$Pr\{t \leq T \leq t + \Delta t\} = R(t) - R(t + \Delta t)$$

And the **conditional probability of failure in the time interval** from t to $(t+\Delta t)$ given the system has survived to time t is

$$Pr(t \leq T \leq t + \Delta t | T \geq t) = \frac{R(t) - R(t + \Delta t)}{R(t)}$$

Then $\frac{R(t)-R(t+\Delta t)}{R(t)\Delta t}$ is the **conditional probability of failure per unit of time** [failure rate, $\lambda(t)$]. Set,

$$\begin{aligned}\lambda(t) &= \lim_{\Delta t \rightarrow 0} \frac{-[R(t + \Delta t) - R(t)]}{\Delta t} \cdot \frac{1}{R(t)} \\ &= \frac{-d(R(t))}{dt} \cdot \frac{1}{R(t)} = \frac{f(t)}{R(t)}\end{aligned}$$

Then $\lambda(t)$ is known as the **instantaneous hazard rate or failure rate function**. Failure rate in some cases may be characterized as increasing (IFR), decreasing (DFR) or constant (CFR) when $\lambda(t)$ is an increasing, decreasing or constant function.

A particular hazard rate function will uniquely determine a reliability function.

$$\lambda(t) = \frac{-d(R(t))}{dt} \cdot \frac{1}{R(t)}$$

$$\lambda(t) \cdot dt = \frac{-d(R(t))}{R(t)}$$

Integrating,

$$\int_0^t \lambda(t) \cdot dt = \int_1^{R(t)} \frac{-d(R(t))}{R(t)} = -\ln(R(t))$$

$$R(t) = \text{Exp} \left[- \int_0^t \lambda(t) \cdot dt \right]$$

Average Failure Rate

The **cumulative failure rate** over a period of time t is defined by

$$L(t) = \int_0^t \lambda(t) \cdot dt$$

A related and useful concept is the **average failure rate** defined between two times t_1 and t_2 .

$$AFR(t_1, t_2) = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} \lambda(t) \cdot dt = \frac{\ln(R(t_1)) - \ln(R(t_2))}{t_2 - t_1}$$

Example: A component has a reliability function given by $R(t) = 1 - \frac{t^2}{a^2}$ for $0 \leq t \leq a$ where a is a parameter of the distribution representing the components maximum life. Give pdf and $\lambda(t)$ of the component. Calculate MTTF, t_{med} and AFR up to the MTTF.